

Worksheet W2: Functions

"Input $\rightarrow$ Machine $\rightarrow$ Output"

Chapter 2 · 12 Problems · Name: _____ Date: _____

Problem 1

Question: What is a **function** in mathematics?

💡 *Hint: Think about a vending machine — input, machine, output.*

Problem 2

Question: What are the **four ways** we describe a function?

💡 *Hint: The "Rule of Four" — V____, N____, V____, A____.*

Problem 3

Question: If the rule is "multiply by 4," what is the output when the input is 5?

💡 *Hint: $5 \times 4 = ?$*

Problem 4

Question: In the function $y = 3x$, what is y when $x = 7$?

💡 *Hint: Multiply x by 3.*

Problem 5

Question: What is the **verbal** rule for the function $y = x + 5$?

💡 *Hint: What operation is being done to x ?*

Problem 6

Question: Complete the table for the function $y = 2x$:

x	0	1	2	3	4
y	0	2	4	_____	_____

💡 *Hint: $y = 2x$. Multiply each x by 2.*

Problem 7

Question: Find the **algebraic** rule for this table:

x	0	1	2	3	4
y	3	5	7	9	11

💡 *Hint: When x goes up by 1, y goes up by 2. The rule has " $2x$ " in it.*

Problem 8

Question: In the function $y = 4x + 2$, what is y when $x = 3$?

💡 *Hint: Multiply x by 4, then add 2.*

Problem 9

Question: What does the "**Mental Zoom**" help you do when reading a graph?

💡 *Hint: What happens to y when x goes up? Is it steep or flat?*

Problem 10

Question: Complete this sentence: "Multiplication is just repeated _____."

💡 *Hint: $3 \times 5 = 3 + 3 + 3 + 3 + 3$.*

Problem 11

Question: A plumber charges $\$50$ per hour plus a $\$30$ call-out fee. Write a function for total cost C in terms of hours h .

💡 *Hint: Total cost = (rate $\times$ hours) + call-out fee.*

Problem 12

Question: If input is 6 and output is 18, and the rule is "multiply by k ," what is the value of k ?

💡 *Hint: $18 = 6 \times k$. Divide both sides by 6.*